Shortest Paths: the Bellman Ford Algorithm

Design and Analysis of Algorithms

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Single-Source Shortest Paths: Algorithms

- Easy cases that we can solve in linear time:
  - Unweighted graphs: use **breadth-first search**.
  - Directed acyclic graphs (DAGs): use **dynamic programming**.
- If edge costs are nonnegative, we'll use **Dijkstra's algorithm**.
- If some edge costs are negative, we use the **Bellman-Ford algorithm**.
- If there is a negative cycle, we are out of luck!

The Bellman-Ford Algorithm

```
c[s] = 0, c[j ≠ s] = Infinity
Repeat n-1 times:
   Tighten every edge in the graph
If c[i] + cost(i, j) < c[j] for any edge (i, j)
   Output "negative cycle detected!"
```

- O(mn) running time.
- Detects negative cycles.
- Easy Analysis: after k iterations of the main loop, each label c[j] reflects the cost of the shortest k-hop path from s to j.

  (Can also interpret as a dynamic programming algorithm)